Homework 4: Project Risk

Team 09

ADA University

Fall 2025 IT Project Management (INFT - 3609 - 10862)

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# 1. Risk Register

The risk register is inspired from the textbook “Information Technology Project Management” which consists of unique ID, rank, risk name, description, category, root cause, triggers, potential responses, risk owner, probability, impact, and status fields. It is designed for “38th International Olympiad in Informatics in Uzbekistan”, authorized on December 20, 2023, and planned to be held during August 9-16, 2026. The total budget is estimated to be $2,817,328 according to previously prepared project budget. In this large project, there are lots of dependencies which requires a continuous risk management plan.

IOI 2019 Azerbaijan host report is used as a lessons learned source and helped to convert the mentioned lessons into risks for IOI 2026 Uzbekistan.

| No. | Rank | Risk | Description | Category | Root Cause | Triggers | Potential Responses | Risk Owner | Probability | Impact | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|
| R08 | 1 | Budget exceed | Since there might be inflation, it can affect the approved total cost resulting scope cuts | Finance | Long timeline; changing markets | Quotes exceed plan; rising burn rate | Sign key contracts earlier; staged procurement; keep extra budget; raise funding decisions earlier | Finance Manager | High | High | Open |
| R01 | 2 | Low speed of network | The contest network may not handle updates immediately and setup quickly causing delays | IT | Not well-designed network | Updates may take longer time; many failures | Review network design; test earlier; have additional equipment; split network into sections | IT Lead | Medium | High | Open |
| R02 | 3 | Translation process delay | Translations may finish late causing delay of printing/preparing of tasks | Operations | Approval dependencies | Translation backlog; last minute changes in texts | Have strict deadlines; daily updates/checks; have extra staff and printers | Translation Lead | Medium | High | Open |
| R06 | 4 | Accommodation shortfall/overbooking | Due to hotel capacity, prices may change, delegations may lack rooms causing extra cost | Logistics | Late booking, high demand | Hotel cancels, terms & conditions change | Book earlier; have backup hotel options; penalties in contracts | Logistics Lead | Medium | High | Open |
| R04 | 5 | System hacked or contest platform goes down | Registration or contest platforms may be targeted, causing downtime and reputational damage | Security | Weak protection | Security test issues; suspicious traffic | Security testing; updates; monitoring backup plan | Security/ IT Lead | Low medium | High | Open |
| R05 | 6 | Server/hardware failure | Hardware may fail due to load during contest week causing delays | Technical | No backup equipment | Errors/warnings during testing | Backup servers; spare parts; fast replacement | IT Lead | Low | High | Open |
| R14 | 7 | Power outage / UPS failure | Power loss may stop contest systems | Facility / IT | No strong power backup plan | Power drops; UPS alarms | Test UPS; backup generator contract | Venue Coordinator & IT Lead | Low | High | Open |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| R03 | 8 | Wrong version prints | Vendor prints outdated or wrong files causing rework | Vendor / Procurement | Poor file control | Proof mismatch; complaints | Single final folder; approval step; contract penalties | Vendor Manager | Low medium | Medium | Open |
| R07 | 9 | Transport delays | Buses may arrive late due to traffic or failures | Logistics | No buffer time or backup buses | Late arrivals | Time buffers; extra buses; alternative routes | Transport Coordinator | Medium | Medium | Open |
| R09 | 10 | Volunteer shortage/weak training | Not enough trained volunteers causing service quality | HR | Late recruitment | Low signups; no-shows | Early recruitment; group trainings; reserve list | Volunteer Coordinator | Medium | Medium | Open |
| R10 | 11 | Medical emergency | Serious health incident may require urgent response affecting event flow | Safety | Limited medical preparation | Many small incidents | On-site medical team; emergency plan | Medical Lead | Low | High | Open |
| R11 | 12 | Venue contract may delay | Late venue approval reduces preparation time | Schedule | Slow process of approvals | Contract not signed | Start contracting earlier; have backup venue | Venue Coordinator | Medium | Medium | Open |
| R12 | 13 | Equipment may arrive late | IT/security equipment delivery may delay causing late testing | Procurement | Late ordering or supplier issues | Missed delivery dates | Early orders; backup suppliers | Procurement Lead | Medium | Medium | Open |
| R13 | 14 | Slow performance of contest platform under load | System slows under heavy usage affecting fairness between competitors | IT | No proper load testing | Failing of load tests | Load testing; performance tuning | QA Engineer | Medium | Medium | Open |
| R15 | 15 | Poor team communication | Teams do not share updates clearly | HR | Unclear process | Conflicting instructions | Weekly meeting; decision log | PMO Coordinator | Medium | Medium | Open |
| R16 | 16 | Sponsor funding may be late | Sponsor payments/ support delayed causing need for extra funding | Finance | Weak agreements/contracts | No signed deals | Have diverse sponsors, backup funding plan | Sponsorship Manager | Medium | Medium | Open |
| R17 | 17 | Key staff may leave | Important people may leave the project causing delays | HR | No backup roles | Resignations | Deputies; documentation; handover plan | Project Manager | Low medium | Medium | Open |
| R18 | 18 | Holidays may slow work progress | Holidays may reduce available workdays | Schedule | Calendar not planned accordingly | Missed internal deadlines | Correct calendar planning | PMO Coordinator | Medium | Low medium | Monitoring |
| R19 | 19 | Permissions may be delayed | Government approvals can take longer than expected | External | Delay of approval process | Extra document requests | Early management | Legal Officer | Low | Medium | Open |
| R20 | 20 | Data privacy issues | Participant data may be handled incorrectly causing trust issues | Compliance | Weak data controls | Audit findings | Access control, privacy checks | Chief of Security | Low | Medium | Open |

# 2. Qualitative analysis

## 2.1. Probability / impact matrix

| Impact / Probability | Low (=1) | Medium (=2) | High (=3) |
|---|---|---|---|
| **High (=3)** | --- | --- | R08 |
| **Medium (=2)** | R18 | R07, R09, R11, R12, R13, R15, R16 | R01, R02, R06 |
| **Low (=1)** | --- | R03, R17, R19, R20 | R04, R05, R10, R14 |

## 2.2. Top ten risks

For qualitative risk analysis, probability/impact matrix has shown. First, for which tasks a deeper analysis is needed was decided. Assessing each risk's probability and impact helped to determine priority. Also, for making the ranking objective numeric scale for the labels (Low = 1, Medium = 2, High = 3) was assigned and calculated a simple risk score by multiplying probability to impact. Then, the risks were sorted to get top ten risks which are R08, R01, R02, R06, R07, R09, R11, R12, R13, R15.

| No. | Rank | Risk Name | Probability | Impact |
|---|---|---|---|---|
| R08 | 1 | Budget exceed | High | High |
| R01 | 2 | Low speed of network | Medium | High |
| R02 | 3 | Translation process delay | Medium | High |
| R06 | 4 | Accommodation shortfall/overbooking | Medium | High |
| R07 | 5 | Transport delays | Medium | Medium |
| R09 | 6 | Volunteer shortage/weak training | Medium | Medium |

| | | | | |
|---|---|---|---|---|
| R11 | 7 | Venue contract may delay | Medium | Medium |
| R12 | 8 | Equipment may arrive late | Medium | Medium |
| R13 | 9 | Slow performance of contest platform under load | Medium | Medium |
| R15 | 10 | Poor team communication | Medium | Medium |

# 3. Quantitative analysis

## 3.1. Explanation of quantitative analysis process

As quantitative analysis requires more effort and numeric assumptions, it is rational to work with top ten risks. In this analysis, those risks were converted into monetary values to understand how much the project can lose if any of these risks may happen. This way, it will be logical to support decisions about contingency reserves. For this, Expected Monetary Value helps to calculate average expected cost of a risk when the probability and cost impact are combined.

First, for each of the risks, an approximate probability percentage is assigned (30%, 35%). Then, these probability values are taken from the qualitative labels (low, medium, high), and improved based on project context and expert judgement.

Second, a single most-likely cost impact is calculated for each risk. Specifically, realistic cost results were included as emergency procurement, rework and so on. In the Microsoft Project, the cost for each task is calculated based on the multiplication of resource rate and work hours with the addition of any fixed costs. As our total cost is $2,817,328, the chosen extra costs should be reasonable and relatively logical to WBS. As a result, EMV is calculated for each task.

## 3.2. EMV estimation table

| No. | Risk Name | Probability | Cost Impact | EMV |
|---|---|---|---|---|
| R08 | Budget exceed | 0.40 | 220 000 | 88 000 |
| R01 | Low speed of network | 0.35 | 180 000 | 63 000 |
| R02 | Translation process delay | 0.30 | 120 000 | 36 000 |
| R06 | Accommodation shortfall/overbooking | 0.25 | 160 000 | 40 000 |
| R07 | Transport delays | 0.30 | 90 000 | 27 000 |
| R09 | Volunteer shortage/weak training | 0.35 | 80 000 | 28 000 |
| R11 | Venue contract may delay | 0.30 | 70 000 | 21 000 |
| R12 | Equipment may arrive late | 0.30 | 90 000 | 27 000 |
| R13 | Slow performance of contest platform under load | 0.25 | 110 000 | 27 500 |
| R15 | Poor team communication | 0.35 | 60 000 | 21 000 |
| | **Total EMV** | | | **3780** |

# 4. Risk response

**R08 - Budget exceed**

The finance manager would manage the risk by tracking the budget each month and comparing it with the planned budget. In case of price increase for hotels or equipment, the most important items will be purchased and contracted earlier for to keep their prices "fixed" and avoid last-minute purchases. Additionally, the big purchases will be split into steps based on priorities, making critical needs secured first. A clear rule will be applied that the reserved extra money can be used only after the approval and only in extreme critical cases. As a last option, non-important things as extra decorations, extra gifts would be reduced or postponed, while contest critical costs are protected.

**R01- Low speed of network**

Before accepting the network design from the supplier, IT team will review and test it in an earlier time for not waiting until the contest week. Computers will be tested and updated mandatorily at the same time in a real environment to see if the network is fast and durable enough. If the test shows negative results as slow speed, the design will be changed earlier and then approved. Furthermore, in overloading and unexpected failures, spare network equipment will be kept on-site and prepared in a way that can be reconfigured quickly. As a last option, switching to a simpler network that is easier and faster to manage will be considered.

**R02 - Translation process delay**

For the contest statements, strict deadlines will be set so that they can't be changed after a certain point. This will help translation process follow a timetable reducing last-minute edits. Progress will be checked daily, and printing will start immediately for each language as soon as it is ready. Additionally, extra printers and staff will be prepared for contest days and peak periods. If delays still appear, a contingency plan will be applied about printing the minimum required materials first. Lastly, a simpler printing process will be considered as a fallback to prevent any further delays.

**R06 - Accommodation shortfall/overbooking**

Hotel rooms will be booked, and contracts will be signed in an earlier time to prevent last-minute cost problems. Moreover, backup hotels will be prepared for preventing capacity problems. The registration numbers will be monitored, and allocations will be updated if demand grows. In case of any warning signs, backup hotel contracts must be activated immediately. As a last option, cheaper options will be considered for staff and volunteers for keeping the main hotel capacity for guests and participants.

**R07 - Transport delays**

Transport routes will be planned and tested at real rush hours beforehand to measure travel time precisely. According to the results, extra timeslots will be added for ceremonies and activities. Additionally, backup plan contacts for extra buses and drivers will be signed for having available vehicles nearby in case of traffic. If delays become serious, there will be adjustments in pickup points. In critical situations, non-important activities will be rescheduled.

**R09 - Volunteer shortage/weak training**

Volunteer recruitment will be started earlier, and a reserve list will be kept for extra volunteers. Trainings will be conducted role-based so that each volunteer would be able to understand their responsibilities clearly. A test day will be conducted before the event to check how volunteers can handle tasks. In the case of low number of volunteers, reserve volunteers will be called. If the problem would continue, staff from lower-priority areas will be reassigned to support operations.

**R11 - Venue contract may delay**

Venue contracting will start earlier and the internal deadlines will be set before the actual deadline. If approvals for contracts would be late, backup venue option will be planned in advance. If there would be delays, only the most critical setup will be prioritized in the first stage and other resources will be added later. As a fallback, the backup venue plan will be used.

**R12 - Equipment may arrive late**

Equipment will be ordered earlier, and suppliers will be tracked weekly. For critical equipment, additional supplier option will be considered in advance to avoid dependence from a single vendor. If there would be late deliveries, the backup supplier will be switched, and express shipping would be considered. For very critical cases, temporary equipment will be rented for not affecting testing phases and contest week.

**R13 - Slow performance of contest platform under load**

Load testing will be done multiple times and problems will be fixed immediately. Extra server capacity will be kept for handling peak usage. If platform still shows slow performance during tests, the contingency plan to increase capacity and simplify configurations will be activated. As a fallback, non-essential features will be disabled, and only core functions for contest will be kept.

**R15 - Poor team communication**

Regular meeting will be held for coordination with written action lists and assigned responsibilities. A decision log will be provided for knowing about the decisions and responsibilities. In the case of risen conflicts, project manager immediately will be informed, and decisions will be made to avoid late problems. As a contingency, extra time will be taken before contest week to do last checks across teams. For the contest week, minimum working plan will be set, and optional things will be postponed for being able to handle the event.

# References


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# Individual Contribution Table

| Team Member | Contribution to the event and report | Estimated % |
|---|---|---|
| Ulkar Ahmadli | Qualitative and quantitative analysis,<br>Risk Response | 50% |
| Javid Ibadov | Risk Register, Risk Response | 50% |
| Raul Hatamov | | 0% |